Brain metastasis

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Abstract | Brain metastasis, which commonly arises in patients with lung cancer, breast cancer and melanoma, is associated with poor survival outcomes and poses distinct clinical challenges. The brain microenvironment, with its unique cell types, anatomical structures, metabolic constraints and immune environment, differs drastically from microenvironments of extracranial lesions, imposing a distinct and profound selective pressure on tumour cells that, in turn, shapes the metastatic process and therapeutic responses. Accordingly, the study of brain metastasis could uncover new therapeutic targets and identify novel treatment approaches to address the unmet clinical need. Moreover, such efforts could provide insight into the biology of primary brain tumours, which face similar challenges to brain metastases of extracranial origin, and vice versa. However, the paucity of robust preclinical models of brain metastasis has severely limited such investigations, underscoring the importance of developing improved experimental models that holistically encompass the metastatic cascade and/or brain microenvironment. In this Viewpoint, we asked four leading experts to provide their opinions on these important aspects of brain metastasis biology and management.

Q *How does the unique brain microenvironment influence brain metastases?*

Adrienne Boire. Cancer represents the pathological 'dark side' of evolutionary biology. A cancer cell is, by definition, genetically unstable, which generates a dynamic and heterogeneous state that bestows the tumour cell with near-infinite adaptability to selective pressure. The microenvironment is the most important source of selective pressure in the system, sculpting the tumour's population dynamics. The brain is wholly unlike any other site of tumour growth in the body and, as such, asserts profound selective pressure on cancer cells.

The brain actually comprises two main microenvironments that differ in their composition — the densely cellular parenchyma and the cerebrospinal fluid (CSF)-filled leptomeninges[1]. The brain parenchyma contains cell types that exist nowhere else in the body, such as astrocytes, microglia, oligodendrocytes and neurons, some of which are electrically active and exist within a vast connectome. Thus, interactions between cancer cells and these brain-specific cell types are unique to brain metastases. In the case of the leptomeninges, the pia and arachnoid contain circulating CSF generated by the choroid plexi, which are structures unique to the brain.

Beyond these anatomical differences, the brain parenchyma and leptomeninges are markedly distinct from the rest of the body at the metabolic level[1]. In the case of the brain parenchyma, neurons take up the majority of oxygen and effectively 'outsource' numerous metabolic tasks to the liver, relying on glucose or ketones to supply sufficient fuel for oxidative metabolism. By contrast, the leptomeninges contain the circulating CSF, which is markedly hypoxic with low concentrations of metabolic intermediates and micronutrients. As cancer cells are quite metabolically active, they must 'solve' these metabolic problems to divide and grow. These metabolic constraints are, therefore, also unique to brain metastases.

These parenchymal and leptomeningeal microenvironments are sufficiently different that they select for distinct characteristics in the metastatic cancer cells that inhabit these spaces[2]. Importantly, these two compartments are anatomically distinct from each other and from the systemic circulation, suggesting that cancer cells within each of these sites possess a distinct array of adaptations and, therefore, vulnerabilities.

Priscilla Brastianos. When metastatic tumour cells arrive in the brain, they encounter a remarkably complex microenvironment that differs from the niche of the primary site. Intracranial metastases need to adapt to this unique milieu to grow effectively. The brain itself consists of a range of cellular constituents, including neurons, astrocytes and microglia, which are not present in the extracranial organs from which the primary cancer originated. Consequently, the metabolic environment in the brain also varies considerably from the primary tumour site; for example, compelling data suggest that breast cancer cells adapt to this environment by adopting 'brain-like' properties, specifically a γ-aminobutyric acid (GABA) ergic phenotype similar to neuronal cells, when they metastasize to the brain[3].

The immune environment is also vastly different in the brain compared with that of other extracranial sites. Historically, the brain has been considered an immune-privileged organ site, whereby immune cell access from the periphery is limited by the blood–brain barrier (BBB). However, data have emerged to suggest that, although the brain might be an immune sanctuary, blood-borne immune cells do in fact enter the brain, especially in the setting of brain metastases. Indeed, the presence of tumour-infiltrating T lymphocytes has now been confirmed in brain metastases from patients with lung cancer, breast cancer, melanoma and renal cell carcinoma[4]. Brain metastases also phenotypically adapt to the immune microenvironment. Notably, expression of cathepsin S, a protease that is usually expressed by leukocytes, was found to be expressed in brain metastases in patients with breast cancer, with preclinical experiments revealing that cathepsin S mediates the transmigration of breast cancer cells through the BBB[5].

Crucially, seminal papers have demonstrated that modulation of the immune system with systemic administration of combined immune checkpoint blockade leads to a clinically

meaningful benefit in ~50% of patients with melanoma that has metastasized to the brain[6,7]. Nevertheless, given that ~50% of treated patients do not benefit from combined immune checkpoint blockade, additional studies are needed to investigate the best approach for modulating the immune system for the optimal treatment of brain metastases.

Livia Garzia. Metastases involving the central nervous system (CNS) can be divided into parenchymal, leptomeningeal and epidural (lesions in the vertebral column) metastases. These microenvironments all offer distinct challenges to and opportunities for circulating tumour cells (CTCs) that travel through the bloodstream, the brain lymphatics or the CSF.

The BBB and the blood–CSF barrier (BCSFB) are the gatekeepers of the CNS, protecting the brain from the potentially lethal consequences of massive inflammation (brain oedema)[8,9], by the action of tight junctions that seal the endothelial cells of brain capillaries (the BBB) or the specialized epithelial and leptomeningeal cells (the BCSFB). These structures render the brain virtually impermeable to hydrophilic solutes, therefore sheltering extravasated and established micrometastases from the action of chemotherapeutics or biologics.

The mechanism that CTCs adopt to cross the BBB and the BCSFB is similar to that of leukocyte transendothelial migration, and brain areas with a low blood flow rate are emerging as preferred sites of parenchymal metastasis[10]. Solitary CTCs that cross the BBB are then challenged to transition from micrometastases to macrometastases. Non-small-cell lung carcinoma cells are more likely to undergo cell death and regression at the micrometastatic stage, whereas for melanoma the most inefficient step is formation of micrometastases[11]. Importantly, within the same cancer type, differences have been highlighted in the alternative occurrence of vascular co-option and neoangiogenesis, which are more prevalent in parenchymal metastases and leptomeningeal masses, respectively[12]. This finding suggests that the subarachnoid space and the brain parenchyma are not equal soils.

Reactive astrocytes, which first come into contact with tumour cells following extravasation, produce plasmin as a defence against metastatic invasion, whereas proteins such as serpins, produced by tumour cells, enable metastatic cells to survive despite plasmin production[13]. Reactive astrocytes infiltrate the progressing brain metastases at the periphery of the lesion, where they influence metastatic progression by relaxing endothelial cell junctions through cytokine secretion and regulate tumour cell gene expression via exosomes[14]. Reactive astrocytes at the periphery of metastases also regulate innate and adaptive immune cells, inducing a switch from a metastasis-hostile brain parenchymal microenvironment to a cancer-promoting one[15].

Microglia can repair the transient damage of the BBB that occurs during CTC transmigration, contributing to shielding of newly formed metastases from systemically administered drugs[16]. When macrometastases are established, microglia and macrophages lose phagocytic activity and adopt an indirect tumour-supportive and anti-inflammatory role through increased anti-inflammatory cytokine production, increased recruitment of peripheral monocytes and inhibition of T cell proliferation. During tumour growth, neurons are subject to damage and cell death due to the persistent activation of microglia, oedema and spatial constraint induced by the physical growth of the tumour mass[17]. On the other hand, however, neural activity fuels high-grade glioma and breast cancer progression[18–20].

Very little is known about the micro-environment of leptomeningeal metastasis. Breast cancer and lung cancer are the two major contributors to leptomeningeal metastasis, but all cancers can spread to this compartment, albeit less frequently. Global studies of the CSF metabolome before and during progression of brain metastasis and, more importantly, of leptomeningeal metastasis are in their infancy[21], but it is conceivable that massive adaptation to growth in suspension and in the absence of the sheltering perivascular niche must occur for tumour cells to successfully colonize the subarachnoid space. Leptomeningeal-tropic breast cancer cells educate their subarachnoid micro-environment by secreting complement component 3 (C3), which, in turn, opens the tight junctions of the epithelial cells of the BCSFB, allowing growth factors to enter the CSF and support metastatic growth[2]. Both the lack of primary human samples for analysis and the inaccessibility of the choroid plexus to in vivo time-lapse imaging have largely hampered our understanding of the initial phases of leptomeningeal colonization, but studies on leptomeningeal metastasis of primary brain tumours have revealed a major role for macrophages in supporting tumour cell proliferation in this compartment[22].

Manuel Valiente. The BBB is an initial and key filter to prevent invasion of metastatic cells into the brain, but it is not the only one. Extracranial organs targeted by metastases share a relatively similar cellular composition with the organ in which the primary tumour originated, inferring a relatively simple process for metastasis-initiating cells to regrow the tumour[23]. By contrast, most

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metastatic cells reaching the brain are not adapted to this microenvironment and, consequently, perish or remain unable to regrow the tumour[11]. The huge selective pressure imposed by the microenvironment is a major contributor in defining which of the initial clones seeding the brain will later succeed during organ colonization and give rise to clinically relevant macrometastases[13,24]. Those rare cells that are equipped with the appropriate 'tools', or that acquire these tools at the moment they are exposed to the brain parenchyma, will survive. However, it is important to note that surviving the action of natural brain defences during the initial moments of colonization does not directly involve the ability to grow in the brain. Surviving metastatic cells might need to evolve in situ to take advantage of specific brain resources that allow them to resume full growth capacity. The process by which cancer cells adapt to this highly specialized microenvironment might involve the selection of additional genomic and epigenomic modifications. Given that the nature of the selective pressure acting on metastatic cells responds to the unique milieu of brain-resident cells, the mechanisms that cancer cells use to overcome this pressure and regrow the tumour in the brain would be, by definition, different from those required at the primary tumour site or even at extracranial metastatic sites.

Consequently, the brain micro-environment is a central aspect that dictates the development of brain metastases. Thus, dissecting the mechanisms required for tumour cells to cross the BBB, block natural brain defences and interact with different components of the brain microenvironment will improve our understanding of key aspects of secondary brain tumour biology, including the specific molecular strategies driving the process of adaptation.

Q *In what ways might studying brain metastases provide insights into primary brain tumours?*

A.B. Primary cancer cells face very similar challenges to their metastatic counterparts — both cell types must face enormous metabolic constraints, evade resident and invading immune cells and continue to grow within the closed CNS. Remarkably, this rather simple observation has enabled the discovery of conserved molecular processes between primary and metastatic brain tumours. For example, the gap junctions originally observed between metastatic cancer cells and astrocytes[25] have also been observed between glioblastoma cells and astrocytes[26] and between diffuse intrinsic pontine glioma cells and astrocytes[19].

A second example is the vasculature supplying the tumour. Studies in both metastatic[27,28] and primary[29] brain tumours suggest that the interactions between cancer cells and blood vessels within the brain are highly heterogeneous, and do not constitute a normal BBB but, rather, a blood–tumour barrier with its own regulatory system. In both of these examples, shared molecular mechanisms between primary and metastatic brain tumours suggest that certain therapeutic approaches might be applicable to both tumour types.

P.B. The rapid advances made in genomic technologies and the recent surge in comprehensive genomic studies have demonstrated that many molecular pathways driving primary brain tumours are also frequently altered in brain metastases. For this reason, the development of therapeutic agents with substantial activity in brain metastases might lead to improved therapies for primary brain tumours, and vice versa.

For example, BRAF and MEK inhibitors demonstrate efficacy in brain metastases from *BRAF*-mutant metastatic melanoma. In 2014, *BRAF*^{V600E} mutations were found to be present in 95% of papillary craniopharyngiomas[30], which are rare brain tumours located close to critical structures (such as the optic chiasm and hypothalamus) and strongly associated with morbidity. Shortly after this genomic discovery, several case reports were published demonstrating dramatic responses to BRAF and MEK inhibitors in patients with papillary craniopharyngiomas[31]. Furthermore, an ongoing phase II trial (Alliance A071601; NCT03224767) is now investigating the efficacy of combined BRAF and MEK inhibition in patients with *BRAF*^{V600E}-mutant papillary craniopharyngiomas, in whom systemic therapy has historically played a limited role. Similarly, oncogenic fusions involving genes encoding receptor tyrosine kinases have been discovered in a broad range of tumour types, including brain metastases and primary brain tumours. Brain-penetrant agents that target fusion proteins (for example, anaplastic lymphoma kinase (ALK) and neurotrophic receptor tyrosine kinase (NTRK) fusions), including entrectinib, have shown high response rates in patients with intracranial metastases in phase I trials[32]. These exciting results are now being extended to primary brain tumours. Advancements in our understanding of brain metastases will certainly advance the field of primary brain tumour research.

L.G. Metastasis-initiating cells or relapse-driving clones are present early on in primary brain tumours (such as medulloblastoma and glioblastoma) and primary lung and breast cancers, indicating that tumour evolution at relapse and metastasis is a branched and intricate non-linear phenomenon in several cancer types[33–35]. Thus, it is conceivable that the tumour microenvironment drives the selection of clones that can successfully repopulate or metastasize and shapes the parallel evolution of cancers of different ontogenies that share a common microenvironment at relapse or metastasis. Beyond the differences in the mechanism of initial tumorigenesis, once tumours are established, glioblastoma and brain metastases from breast cancer and lung cancer share many similarities. For example, the extensive interaction between tumour cells and macrophages or microglia in the tumour microenvironment occurs in both brain metastases and subgroups of primary glioblastoma[36], and issues of drug delivery across the BBB are shared between different entities that colonize the brain and primary brain tumours[37]. One of the major challenges in our understanding of the biology of primary and secondary brain tumours is the paucity of human samples available for molecular profiling during the course of disease progression, so there is a clear advantage in grouping brain cancers according to their molecular driver genes (for example, *BRAF* or *PTEN*/*PIK3CA*/*MTOR*) and patterns of invasion and/or metastasis (for example, parenchymal versus leptomeningeal), as this strategy will reveal patterns of convergent evolution of different cancers that colonize similar microenvironments. Leptomeningeal brain metastases of breast cancer also show a convergence with medulloblastoma leptomeningeal metastases, whereby they both harbour upregulated expression of CC-chemokine ligand 2 (CCL2), which potentially mediates extravasation of CTCs[2,38,39].

M.V. Despite what one could imagine, there are many differences between brain metastases and primary brain tumours.

The most remarkable difference is that glioblastoma, the most frequent primary brain tumour (but much less prevalent than brain metastases), rarely metastasizes outside the brain despite its strong invasive

nature. Although the explanation for this phenomenon remains unknown, it could have important implications. By contrast, brain metastases are commonly found in patients who have additional extracranial systemic metastases, which seems to be important for response to immunotherapies[40]. Consequently, studying the crosstalk between brain metastases and the immune system might generate new opportunities to increase the sensitivity of primary brain tumours to immunotherapy.

There are also therapies that are efficacious in primary brain tumours but not in brain metastases. Indeed, clinical trials with the alkylating agent temozolomide in patients with brain metastasis have not been encouraging[41]. However, as glioblastoma responses to temozolomide are transient and every patient eventually develops resistance, dissecting the lack of sensitivity in brain metastases at the molecular level might offer new strategies to challenge glioblastoma relapse. This approach is especially interesting given that preclinical evidence has shown that temozolomide could be effective in a preventive scenario for brain metastases[42], suggesting that therapeutic sensitivity could be modulated in cancer cells.

In addition to therapeutic response, there are also clear differences regarding the underlying biology. Notably, glioblastoma has a more potent pro-angiogenic nature than brain metastases[43], and further research on this topic might inspire the development of imaging approaches to non-invasively differentiate between primary brain tumours and brain metastases with high specificity. There are also important similarities. Although glioblastoma is more invasive than an established brain metastasis, which remains well circumscribed with discrete invasive fronts, the basic mechanism of local invasion is similar. Glioma cells and metastatic cells in the brain use pre-existing blood vessels to disseminate locally and increase their resistance to various stressors[44]. Comparative analysis of the underlying regulatory mechanisms of this process of vascular co-option could identify common therapeutic targets for primary and secondary brain tumours. Additional common vulnerabilities in primary and secondary brain tumours include metabolic pathways[45]. The fact that metastatic cells need to alter their metabolism when growing in the brain also infers that the initial metabolic profile present in the primary tumour might be incompatible with survival in the brain[45,46]. Thus, reverting the metabolic state acquired in the brain could be a new therapeutic strategy, not only for brain metastasis but also for primary brain tumours.

Increasing the number of comparative studies between primary brain tumours and brain metastases could offer a unique vision of the underlying biology of brain malignancies and generate unpredicted therapeutic opportunities. These comparative approaches will eventually reach well-known clinical scenarios for which there are currently no explanations, such as the higher incidence of epilepsy in patients with primary brain tumours than in those with brain metastases.

Q *How can we improve preclinical models of brain metastasis?*

A.B. Preclinical models of brain metastasis are just that — models of human disease. There are no 'bad' models, only models that have simply been ill-used or findings that have perhaps been over-interpreted. The 'best' model for the study of brain metastasis really depends on the question that is being asked. If the question is related to cancer cell–immune cell interactions, then a syngeneic or immunocompetent system is ideal. If the question focuses on progression of primary tumours to metastases, then a genetically engineered mouse model (GEMM) is ideal. If the question relates to tumour heterogeneity or human-specific genetic changes, xenograft systems are often the best option.

Importantly, as a community, we are undertaking these lines of inquiry in the 'post-omics' era. Improved sequencing capacity for DNA, RNA and protein (and soon metabolites) at the bulk and single-cell levels has been joined with corresponding revolutions in computational biology, such that we now have the capacity to cope with the frankly enormous data sets generated from human samples. Collaborative projects such as the Human Tumor Atlas Network (HTAN) will become foundational for our field. In the near future, we may well ask ourselves whether we should be using mouse models as our primary tool for discovery and what kinds of questions we can reasonably expect to answer with them.

P.B. The metastatic cascade to the brain involves growth within the primary tumour, intravasation into the bloodstream, survival within the circulation, extravasation and invasion into the brain, and angiogenesis and proliferation within the brain parenchyma. Preclinical models attempt to recapitulate these steps within the cascade. In vitro assays can be used to study the passage of cancer cells through the BBB[47]. In vivo, cancer cells implanted intracranially (via stereotactic injection) in murine models allow investigators to monitor growth within the brain parenchyma, and fluorescently labelled cells can be tracked using bioluminescent imaging[48]. These models, although extremely helpful for studying specific aspects of metastasis, are limited in that they only mimic one part of the metastatic cascade. Intracarotid and intracardiac injection of cancer cells can model dissemination of the cancer cells into the bloodstream and can be used to study extravasation through the BBB as well as colonization and growth within the brain. Kienast et al.[11] employed an intracranial window in nude mice to track individual fluorescently labelled cancer cells injected via the carotid artery; this model reflects colonization into the brain, including the interaction of the metastases with the vasculature of the brain. The vast majority of existing mouse models of brain metastases use immunodeficient mice and human cell lines, in which the lack of an immune system limits the ability to study the interaction of cancer cells with the immune microenvironment. Syngeneic mouse cell lines with brain tropism are also used, although many of the immunocompetent mice develop a high burden of extracranial metastases when these cell lines are injected into the bloodstream. Few GEMMs of brain metastases that recapitulate the metastatic cascade have been developed, but those few that have been generated often demonstrate a variable frequency of extracranial and intracranial metastases and might not be representative of the exact genetic context of human brain metastases[49–51]. Improving preclinical models of brain metastases will expand our ability to study the biology of brain metastases and, more importantly, identify novel therapies for this disease. A concerted effort from the scientific community is needed to expand the number of cell lines available with brain tropism, as well as to optimize models to better mimic the complex metastatic cascade.

L.G. Spontaneous (transgenic) or syngeneic models of brain metastasis are the best models to study the interaction between brain metastases and the brain microenvironment. However, one drawback of using transgenic models for the formulation of new hypotheses (when human samples are not available) or validation of candidate driver processes in brain metastases is their relative genetic

simplicity; transgenic models have a clonal representation of few selected mutations, whereas human metastases show a relevant subclonal heterogeneity that influences their evolution during progression and metastasis. To this end, incorporating steps that create intratumour heterogeneity by random somatic mutagenesis with viruses or transposons will improve the predictive value of the model for metastasis[52,53]. Transplantation-based preclinical models such as patient-derived xenografts — which are preferred over xenografts established from highly passaged cell lines — have the advantage of recapitulating human genetic and epigenetic alterations and intratumour heterogeneity, but at the cost of an intact immune system in the immunocompromised rodent host. The recent development of immunocompetent humanized recipient mice for establishing xenografts of human origin seems to be a promising avenue to improve preclinical models of brain metastasis, although more research is needed to completely evaluate their limitations, especially in long-term studies in which progressive loss of human immune cells might be an issue[54]. When developing a brain metastasis preventive approach or when studying the early steps of metastatic colonization to the brain, it is crucial that patient-derived xenograft models are implanted orthotopically in order to mimic as closely as possible the entirety of the metastatic cascade. As the vast majority of biologics and targeted therapies for brain metastasis will be in the context of heavily treated patients, a further optimization step of current preclinical models is to include therapeutic protocols that recapitulate the clinical scenario, as discoveries made in untreated preclinical models could be of limited predictive value when translated to heavily treated patients.

M.V. The ideal in vivo model of brain metastasis would completely recapitulate the metastatic cascade, have an intact immune system that co-evolves during cancer progression and be compatible with therapeutic interventions (for example, surgery, radiotherapy and chemotherapy). Despite the large number of GEMMs available for lung cancer, breast cancer and melanoma (which are the main sources of brain metastasis), only three have been shown to give rise to brain metastasis[49–51]. These GEMMs are infrequently used by brain metastasis investigators, the reasons for which might be related to the low and variable incidence of brain metastases, the lack of macrometastases derived from the primary tumour or extracranial metastases (due to early death), the absence of reporters that are compatible with non-invasive high-throughput imaging in cancer cells and the differences between the molecular profile of some of these experimental models and the human disease. It is also true that many other commonly used GEMMs have not been characterized for their incidence of brain metastasis. To cite two examples, the *Kras*^{G12D};*Trp53*-null lung adenocarcinoma[55] and ERBB2-activated breast cancer[56] GEMMs did not interrogate the presence of brain metastases, although lung cancer and breast cancer are among the cancer types with the highest rate of brain metastasis[1]. It seems appropriate to extend the characterization of these, and new, GEMMs to the brain in addition to incorporating systemic therapies that could increase the time available for brain metastasis to develop.

Alternatively, given the available information on molecular alterations present in brain metastases, which differ from those in the primary tumour and other extracranial metastasis, it is tempting to speculate whether engineering such alterations in GEMMs could be a valuable strategy to generate improved models. In favour of this approach, a genetic manipulation of *Akt1* introduced into a spontaneous melanoma mouse model correlated with an impressive incidence of brain metastases[51]. Given the extensive data demonstrating the functional importance of the AKT pathway in human brain metastasis, this study could be considered the proof of concept for this approach. However, the genetic *Akt1* alteration introduced in mice is not prevalent in the human disease, and the recent engineering of an *Akt1* mutation that is prevalent in human melanoma in mouse primary tumours was shown to provide general pro-metastatic attributes favouring a pan-metastatic phenotype rather than brain-specific metastasis[57]. In fact, no highly abundant and recurrent mutations are associated with brain metastasis in humans, which imposes a challenge for deciding which alteration should be prioritized in models. Gain-of-function screens containing clinically relevant candidates could perhaps be a solid strategy for development of GEMMs that metastasize to the brain.

Spontaneous models of brain metastasis from a primary tumour are highly valuable for developing biomarkers or therapeutic strategies to prevent brain metastasis. However, whether they are strictly necessary to understand the biology of brain colonization and develop strategies aimed at impairing the viability of cancer cells once they have reached the brain is a matter of continuous debate. By contrast, the most popular experimental approach consists of injecting brain-tropic cell lines into the systemic circulation to induce metastasis in the brain. In favour of these models, there are publications validating the same mechanisms using brain-metastatic (BrM) cell lines and GEMMs that develop brain metastasis[58]. In addition, work with BrM cell lines has provided new therapeutic opportunities that have been translated to patients, such as the inhibition of gap junctions between cancer cells and astrocytes or the inhibition of signal transducer and activator of transcription 3 (STAT3) in the same glial cell[15,25].

Alternative models for exploratory purposes in which multiple candidates can be tested do exist. One of the most popular strategies is brain organotypic cultures, which consist of an ex vivo approach to interrogate either early or advanced stages of brain colonization using genetic or pharmacological strategies[13,15].

Q *What are the unique therapeutic targets in the brain and how can we improve treatment of brain metastases?*

A.B. There are two major categories of treatments tailored to the unique therapeutic targets in the brain — anti-neoplastic agents that overcome the BBB and the BCSFB, and targeted therapies that interrupt cancer cell–microenvironment interactions.

In the first case, understanding these barrier systems in states of both health and malignancy (that is, both the BBB and the blood–tumour barrier) is essential. Our understanding of these barrier systems in health is excellent. However, we are only just beginning to understand the function and regulation of the blood–tumour barrier. I hasten to add that the relationship between the parenchyma and the leptomeningeal space is far from completely understood. It is clear that the relationship between these compartments is complex, and is currently best captured by the 'glymphatic' hypothesis, which posits that CSF passes around the perivascular spaces, enters the parenchyma, where it collects solutes from the extracellular fluid, and is then collected around the perivascular spaces, where it eventually joins the venous system and/or the newly uncovered lymphatics[59–61]. At present, when studying a novel compound, it is essential that we concurrently collect blood, tumour tissue and CSF for pharmacokinetic analyses.

Too often, CSF is collected as a proxy for drug delivery into the brain parenchyma.

In the second case, targeting cancer cell–microenvironment interactions in clinical practice is extraordinarily challenging. As patients with brain metastases very often harbour disease outside the brain, both intracranial and extracranial disease must be addressed. Clinical trials designed to presuppose a population with active disease exclusively within the brain will result in slow accrual and will not reflect current practice. However, trials allowing enrolment of patients with active extracranial disease must be designed and interpreted with care. Consensus opinions, in particular those from the Response Assessment in Neuro-Oncology (RANO) collaborative group, have provided a roadmap to aid clinical investigators in addressing these issues[62].

Ultimately, I foresee that strategies targeted to brain metastases (for example, anti-connexin 43 antibodies) will be joined with tumour-directed therapies that can penetrate the brain and CSF (for example, the epidermal growth factor receptor (EGFR) inhibitor osimertinib). Ideally, this approach of combining multiple orthogonal strategies will eliminate toxic untargeted treatments such as radiotherapy, thereby both increasing the survival and improving the quality of life of patients with brain metastases.

P.B. Clinically, patients often have divergent therapeutic responses in intracranial and extracranial lesions. Following progression of the brain lesions, >50% of patients die from their brain metastases. The BBB and the blood–tumour barrier play crucial roles in the decreased intracranial response rates that are often seen clinically with systemic therapies[27]. Whether genetic heterogeneity contributes to this divergence in therapeutic response in brain metastases remains an open question. In breast cancer, data have emerged showing shifts in the expression status of hormone receptors and ERBB2 between intracranial and extracranial metastases[63,64]. In another study, genomic characterization of 86 matched brain metastases, primary tumours and normal tissues demonstrated branched evolution, whereby brain metastases and primary tumours shared a common ancestor, yet the brain metastases and primary tumours diverged genomically[65]. As a result of this branched evolution, brain metastases harboured genetic alterations that were undetected in the primary tumour, including driver mutations for which targeted therapies are available. Clinically, this finding implies that genetic screening of the primary tumour, which is often performed as the standard of care in the clinic, might miss potentially actionable targets in brain metastases. Signalling pathways commonly altered in brain metastases include the PI3K, cyclin-dependent kinase (CDK) and EGFR pathways; compared with matched primary tumours from the same patient, many of these genetic alterations are exclusive to the brain metastasis[65]. A genomically guided phase II trial (Alliance A071701; NCT03994796) has now been initiated, in which intracranial and extracranial tissue from patients with progressive brain metastases will be sequenced for specific genetic mutations (CDK pathway, PI3K pathway and NTRK and/or ROS fusions), and patients will be treated with the corresponding CNS-penetrant inhibitor (abemaciclib, GDC-0084 or entrectinib). The objective of this unique precision medicine study is to determine whether targeting alterations observed in brain metastases will improve clinical outcomes.

L.G. The presence of the BBB and the BCSFB constitutes a unique opportunity to develop targeted strategies to prevent metastasis by blocking the mechanisms that mediate extravasation. Several attractive targets have been identified and are the subject of intensive study in parenchymal brain metastasis[5,13,47], whereas the identification of key factors for BCSFB transmigration in leptomeningeal disease is lagging behind, with very few targets identified[53]. Reactive astrocytes also offer a unique opportunity to manage brain metastasis, with STAT3 inhibitors at the forefront[15]. Modulators of neural activity, which have been targets of drug development to cure neurological conditions, are an untapped arsenal of compounds with favourable brain distribution that could be repurposed once the extent of crosstalk between brain metastases and neurons is further investigated. Given their role in primary and metastatic brain tumours, inhibitors of neurotransmitter receptors (such as the GABA receptor) and metabolism[66], as well as agents targeting brain-specific growth factor signalling or metabolism[67], represent attractive targets. The composition of the extracellular matrix of brain tumours is also unique and could be targeted by biologics in the context of combination therapies. Indeed, targeting tenascin C and hyaluronic acid, which are abundant in the brain extracellular matrix, has shown promise in preclinical studies[68]. Exploiting the unique microenvironment of brain metastasis with immunotherapy approaches in combination with targeted agents directed against molecular drivers present at the time of treatment will have the greatest impact on patient therapies. In light of our deeper understanding of tumour evolution, a key aspect in the management of brain metastasis will be to promote the use of tissue biopsies to confirm the presence of the target when it is safe to do so, and also to prioritize agents that were efficacious in preclinical models closely mimicking patient care, especially when targeted agents are to be used in an adjuvant or second-line treatment scenario.

M.V. The pressure imposed on cancer cells by the microenvironment forces them to adapt or perish. Two main aspects of this biology have been identified and validated as potential anti-metastasis strategies — the anti-tumorigenic behaviour of the naïve brain and the pro-metastatic niche provided by the rewired tumour-associated microenvironment. The rules governing the crosstalk between cancer cells and the microenvironment are derived from the unique cellular compartments present in the brain. Astrocytes and microglia have been shown to have the ability to kill cancer cells[13,24]; however, these defences could be circumvented by different molecular mechanisms hijacked by metastasis-initiating cells. Targeting these mechanisms or reinforcing the anti-tumorigenic nature of glial cells could be a valuable strategy to be explored. An additional strategy taken by cancer cells is to reprogram the brain microenvironment. Indeed, the anti-tumorigenic nature of the brain is less evident once metastases have grown. Interestingly, this finding correlates with the activation of signalling pathways that are not present in the naïve brain microenvironment[15,16]. Targeting these pathways terminates the pro-tumorigenic behaviour of the rewired brain microenvironment[5,15], but might also boost the anti-tumorigenic nature of the brain.

In addition to the unique targets present in the brain microenvironment, cancer cells themselves also respond to the new microenvironment by altering gene and protein expression patterns in situ[20,69]. Emerging evidence suggests that a new avenue for therapeutic intervention might be derived from the specific vulnerabilities created in the process of adapting to the brain microenvironment[58].

An area that remains unexplored given the emergency imposed by the limited survival of patients is the influence of brain

metastasis on neurocognition. Although neurocognitive impairment is one of the most frequent brain-specific symptoms among patients with brain metastases, very limited research has been devoted to this at the experimental level. Recent findings in primary[19,26] and secondary[20] brain tumours demonstrating that cancer cells integrate into brain circuits to hijack the synaptic machinery have opened the possibility of dissecting their impact on neuronal processing. Improvements in patient survival outcomes will increase the importance of quality of life and it will become more important to understand how metastatic cells influence neurons at the cellular level and also neural circuits. These questions will require new expertise and new models to explore a very unique aspect of metastases that grow in this organ.

In summary, treating brain metastases using the same systemic therapy given to the primary tumour has proved ineffective (with some exceptions for very small groups of patients; for example, ALK inhibitors), and the brain microenvironment is emerging as a potential resource to improve therapies directed against brain metastasis. These considerations suggest that it might be appropriate to consider organ-specific therapies. Thus, it is essential that preclinical efforts take this into account. In this sense, performing subcutaneous injections (xenograft models) to test anti-metastasis compounds is never the best option. Similarly, it is important to differentiate preventive experimental treatments from interventional ones that act on established metastases, which are closer to the clinical situation and include a rewired pro-tumorigenic brain microenvironment.

In addition to developing new therapeutic strategies, it is also very important to understand why available treatments fail. For instance, relapse after surgery has not been modelled in mice, but it is expected that interactions with the brain microenvironment are as important after surgery as they are during the initial colonization of the brain (such as, for instance, vascular co-option)[13,15]. In addition to relapse, the lack of efficacy of whole-brain radiotherapy suggests that clinicians should move away from this approach in favour of more localized treatments, such as stereotactic radiosurgery. However, the reasons for these poor responses to radiotherapy in the context of brain metastasis have barely been addressed experimentally. Investing additional research into the molecular mechanisms of resistance will improve our knowledge of brain metastasis biology and also offer ways to improve available therapies.

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Competing interests

A.B. is a Consultant for Arix Biosciences, and has patents related to methods for treating brain metastasis (US Patent 10,413,522) and modulating permeability of the blood–CSF barrier (US Provisional Application Number 62/258,044). P.B. has consulted for Lilly, Genentech–Roche, AngioChem and Tesaro, has received speakers honoraria from Merck and Genentech–Roche, and has received research funding from Merck, Bristol-Myers Squibb and Pfizer. The other authors declare no competing interests.

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